

# Mike Ichikawa

DATA SCIENTIST · ML ENGINEER · DATA ENGINEER

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## SUMMARY

Founder & Engineer at Microclaw LLC, building a production AI agent for Microsoft 365 submitted to the Microsoft Commercial Marketplace. KNN tool selection, RAG semantic memory retrieval, smart model routing, in-house eval framework. Previously nine years teaching undergraduate mathematics at Santa Rosa Junior College and three years semiconductor physical design at Intel. MS Mathematics (CCNY, 2014), BS Mechanical Engineering (UC Berkeley). Portfolio covers time series forecasting, anomaly detection, NLP, cloud ETL, ML serving, a Databricks medallion lakehouse, C++ performance extensions via pybind11, and a five-repo paper trading system.

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## TECHNICAL SKILLS

|                  |                                                                                                                               |
|------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Languages        | Python · SQL · C++ · R · TypeScript · Node.js                                                                                 |
| ML / Statistics  | scikit-learn · Prophet · PyTorch · XGBoost · LightGBM · scipy · statsmodels · Bayesian inference                              |
| LLM / NLP        | Anthropic API · Azure OpenAI · Microsoft Graph · MSAL · HuggingFace · FinBERT · prompt engineering                            |
| Data Engineering | PostgreSQL · Databricks · Delta Lake · medallion · dbt · SQLAlchemy · pandas                                                  |
| Cloud / Infra    | AWS (S3 · Lambda · DynamoDB · CloudWatch) · Docker · FastAPI · Redis · Pydantic v2 · boto3 · Microsoft Commercial Marketplace |
| Tools            | Git · pytest · Vitest · Jupyter · Streamlit · Docker Compose                                                                  |

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## EXPERIENCE

### Founder & Engineer

2026 – present

Microclaw LLC · Portland, OR

- Built a production AI agent on Microsoft Teams (Azure OpenAI GPT-4o + function-calling) operating across 12 Microsoft 365 services through 85 tool functions on Microsoft Graph; submitted to Microsoft AppSource (Standard + Self-Hosted offerings)
- Implemented KNN tool selection over 1,020 synthetic training examples using `text-embedding-3-small` (k=7, 0.80 confidence threshold, cosine-similarity fallback): 95.0% recall vs 92.7% embedding-only baseline, 6.4 tools selected per request vs 18, ~60% input-token reduction
- Designed RAG-pattern semantic memory retrieval (cosine similarity over per-user persistent context) and a smart model router (GPT-4o-mini default, escalation to GPT-4o on complexity signals; 17× cheaper for simple queries); built an in-house eval framework scoring agent output across 75 tasks against a baseline bot
- Built a natural-language automations layer where users describe IFTTT-style rules in plain English; the agent parses intent into structured rules persisted to Azure Postgres, fired in real-time via Microsoft Graph change-notification webhooks. 290 Vitest unit tests; Bicep IaC; Docker → GHCR release pipeline via GitHub Actions

### Mathematics Faculty

2015 – 2023

Santa Rosa Junior College · Santa Rosa, CA

- Taught undergraduate mathematics for nine years: Algebra through Calculus III, Linear Algebra, Differential Equations, Statistics
- Designed assessments and iterated course materials from cohort performance data
- Taught STEM majors alongside non-STEM majors fulfilling general-education math requirements at an open-enrollment community college

- Microprocessor physical design on Intel Ivy Bridge (22nm), including timing closure, power analysis, and design rule verification
- Built Python and Perl automation for physical verification workflows, processing millions of layout data points per run

## EDUCATION

|                                  |                                    |      |
|----------------------------------|------------------------------------|------|
| <b>MS Mathematics</b>            | The City College of New York       | 2014 |
| <b>BS Mechanical Engineering</b> | University of California, Berkeley | 2008 |

## PORTFOLIO PROJECTS — full portfolio at [mtichikawa.github.io](https://mtichikawa.github.io)

### Trading System Arc Live Demo · [github.com/mtichikawa](https://github.com/mtichikawa) — 5 repos

Five interconnected repositories: live OHLCV pipeline (ccxt/Kraken + PostgreSQL) → candlestick chart generation (mplfinance) → dual-path signal engine (EMA/RSI/MACD/BB + local FinBERT sentiment, 0.6/0.4 fusion) → parameter-optimizing backtester (Sharpe/Sortino/drawdown, staged grid search, T3 feedback loop) → Streamlit multi-page oversight dashboard with live signal display, equity curve, and parameter review. 174/174 tests across T2–T5.

[ccxt](#) · [PostgreSQL](#) · [mplfinance](#) · [FinBERT](#) · [pandas](#) · [NumPy](#) · [Streamlit](#) · [Plotly](#) · [pytest](#)

### Databricks Medallion Lakehouse [github.com/mtichikawa/databricks-lakehouse](https://github.com/mtichikawa/databricks-lakehouse)

Medallion lakehouse (bronze → silver → gold) on 10M NYC taxi rows with Delta Lake. Append-only bronze with provenance metadata; silver types, dedupes, and quarantines invalid rows rather than dropping them; gold serves query-specific aggregates. Row-level data quality gates between layers halt the pipeline on schema drift or null-rate breaches. 95 tests across bronze/silver/gold/quality/pipeline. Runs locally via deltalake; deployable to Databricks Community Edition.

[Delta Lake](#) · [medallion architecture](#) · [pandas](#) · [pyarrow](#) · [data quality gates](#) · [pytest](#)

### Real-Time Anomaly Detection [github.com/mtichikawa/anomaly-detection](https://github.com/mtichikawa/anomaly-detection)

Streaming ensemble detector: Isolation Forest, LSTM autoencoder, and Z-score with 2-of-3 majority voting. Dockerized FastAPI REST API with health checks and configurable detector selection. Live interactive demo on portfolio site.

[Docker](#) · [FastAPI](#) · [scikit-learn](#) · [PyTorch](#) · [Redis](#) · [pytest](#)

### Cloud ETL Pipeline [github.com/mtichikawa/cloud-etl-pipeline](https://github.com/mtichikawa/cloud-etl-pipeline)

AWS-native ETL: tenacity-based REST API ingestion with exponential backoff → Parquet/Snappy storage on S3 with Hive date partitioning → Lambda validation → DynamoDB run logging for failure traceability. Tests mock all AWS calls with moto.

[AWS S3](#) · [Lambda](#) · [DynamoDB](#) · [boto3](#) · [pandas](#) · [Apache Parquet](#) · [moto](#)

### SQL Analytics Pipeline [github.com/mtichikawa/sql-analytics-pipeline](https://github.com/mtichikawa/sql-analytics-pipeline)

Three-layer dbt-style pipeline (staging → clean → marts) on NYC Airbnb data. Window functions, CTEs, SQLAlchemy ORM with swappable SQLite/PostgreSQL backend. Layered transform pattern enforces raw data immutability.

[PostgreSQL](#) · [SQLAlchemy](#) · [dbt-style transforms](#) · [SQLite](#) · [pandas](#)

### LLM Data Analysis Assistant [github.com/mtichikawa/llm-data-assistant](https://github.com/mtichikawa/llm-data-assistant)

Two-path query routing: rule-based fast path handles ~70% of queries (correlations, aggregations, group-bys) with zero API cost. Queries below confidence threshold escalate to Anthropic API with DataFrame context injection and multi-turn conversation history.

[Anthropic API](#) · [pandas](#) · [hybrid routing](#) · [multi-turn conversation](#)

*Additional: Multi-Armed Bandit A/B Testing (Thompson Sampling · UCB1 · Bayesian inference) · Bias Detection in LLMs (ANOVA · Cohen's d) · Financial NLP Parser (SEC EDGAR · regex · sentiment lexicon) · GitHub Trend Forecaster (Prophet · PyTorch LSTM in progress)*